

INTERNATIONAL GCSE

BIOLOGY

9201/2

Paper 2

Mark Scheme

June 2024

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from [oxfordaqa.com](https://www.oxfordaqa.com)

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

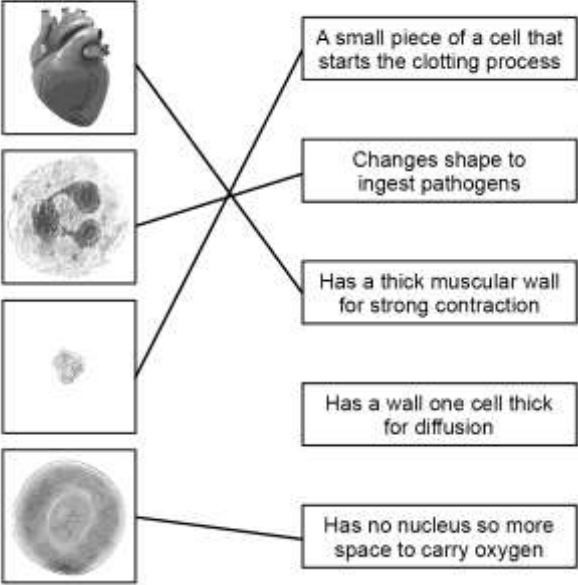
Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	Structure in the circulatory system  Description - A small piece of a cell that starts the clotting process - Changes shape to ingest pathogens - Has a thick muscular wall for strong contraction - Has a wall one cell thick for diffusion - Has no nucleus so more space to carry oxygen Do not accept more than one line from a box on the left		4	AO3 3.2.3b,l,n, p, q 3.4.7c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	4 (days)		1	AO2 3.2.3, o 3.4.7c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	any two from: (white blood cells) • increase before symptoms start or increased just after the child is infected • increased then decreased • maximum on day 4 or maximum 3 days after infection or maximum 2 days after symptoms start • still high when symptoms stop	allow maximum at 16 000	2	AO3 3.2.3o 3.4.7c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	protein		1	AO1 3.2.3r

Total			8	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	All correct = 3 marks, 2 or 3 correct = 2 marks, 1 correct = 1 mark do not accept more than one line from a box on the left Level of organisation Description		3	AO1 3.1.2 b, c, d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	white blood cell blood heart circulatory system	allow 1 mark for wbc → blood allow 1 mark for blood → heart allow 1 mark for heart → circulatory system	3	AO2 3.1.2b, c, d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	differentiation	ignore specialisation	1	AO1 3.1.2a 3.1.1d 3.5.2i,k

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	any three from: - nucleus to contain DNA / genes / chromosomes (to fertilise the egg) (many) mitochondria to release energy for movement / contraction (long) tail to swim / lash (to reach the egg) - enzymes to penetrate the (outer layer) of the egg - streamlined shape to allow faster movement	ignore genetic material / information do not accept produce / make / create energy	3	AO2 3.1.2a 3.1.1d

Total			10	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	infection (in person or animal)	allow named infection do not accept a viral, protist or fungal infection	1	AO3 3.4.7 f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	household / hospitals / industrial cleaning	allow examples such as cleaning a kitchen work surface	1	AO3 3.4.7

Question	Answers	Mark	AO/ Spec. Ref.
03.3	Level 3: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO4
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the method is not fully logically sequenced.	3–4	AO4
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	AO4
	No relevant content	0	
	Indicative content (method) • prepare sterile nutrient agar plates • add lawn of bacteria • prepare wells or discs • add disinfectants A, B and C to separate wells / discs • incubate between 20 °C and 30 °C • for suitable time (12–48 hours) • observe / measure diameter of clear zone or • prepare test tubes of sterile nutrient broth • add culture of bacteria • count number of bacteria present • add disinfectants A, B and C to separate tubes • incubate between 20 °C and 30 °C • for 3–24 hours • count number of live bacteria remaining		3.4.7 6.2.5

	(aseptic technique) • use aseptic technique to avoid contamination • examples of aseptic technique appropriate to method chosen • working near a Bunsen burner • sterilise loop in flame • autoclave Petri dish to sterilise • flame the necks of bottles / tubes (control variables) • control variables relevant to method chosen • volume of disinfectant • concentration of disinfectant • volume of nutrient broth • volume of bacteria culture • time of incubation • temperature of incubation between 20 °C and 30 °C For Level 3 the measurements, aseptic technique and control variables must match the method chosen.		
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Total		8
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	4(.0) (arbitrary units)		1	AO2 3.4.5 6.3.11

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	(blood glucose concentration) decreases then increases	allow falls then rises	1	AO2 3.4.5 6.3.11
	(decreases) from 8 to 3.5 (au) or (increases) from 3.5 to 4.2 (au)	allow decreases by 4.5 (au) allow decreases until 09:00	1	
		allow increases after 9:00 by 0.7 (au) allow increases after 9:00 until 09:30 allow tolerance of $\pm$ half a small square for 3.5 and 4.2 only		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	increased blood glucose concentration causes release of insulin (from the pancreas)		1	AO3 3.4.5a,b, c
	(so) (liver / muscle) cells absorb glucose (from the blood)		1	AO1
	glucose is converted to glycogen and stored (in the liver / muscles)		1	AO1
	(then) blood glucose concentration falls too low so glucagon is released (from the pancreas)		1	AO3
	(which) stimulates the conversion / breakdown of glycogen to glucose		1	AO1
	(glucose is) released into the blood so blood glucose concentration rises		1	AO3
Total			9	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	both animals and plants		1	AO1 3.2.6d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	both day and night		1	AO1 3.2.6d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	mitochondria		1	AO1 3.2.6e 3.1.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4		max 3 marks if answer is not comparative		
		answer only needs to refer to more / faster once		
	to get more oxygen / air into the lungs		1	AO2 3.2.6 b,f g
	(for) increased (rate of) uptake / diffusion of <u>oxygen</u> into the blood / muscle		1	AO2
	(for) faster / more respiration in muscle (cells)		1	AO1
	to release more energy for movement / swimming / contraction	do not accept energy produced / made / created	1	AO1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	not enough oxygen for aerobic respiration		1	AO2 3.2.6 i, j, l
	(so increased) anaerobic respiration produces lactic acid		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	(increase =) 11.3	allow 12.5 – 1.2	1	AO2 3.2.6 i, j, l 6.3.3
	(% increase =) $\frac{11.3}{1.2} \times 100$		1	
	= 941.66(666....)	allow a correct answer using incorrect values from the table	1	
	= 941.7		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.7	any one from: • muscle fatigue • (muscle) cramp • less efficient muscle contraction		1	AO2 3.2.6l

Total		14
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	length of the body		1	AO2 3.6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.																				
06.2	<table><tr><td></td><td colspan="3">Cause of variation</td></tr><tr><td>Characteristic</td><td>Genes only</td><td>Environment only</td><td>Both genes and environment</td></tr><tr><td>Break in the shell</td><td></td><td>✓</td><td></td></tr><tr><td>Length of the body</td><td></td><td></td><td>✓</td></tr><tr><td>Number of bands on shell</td><td>✓</td><td></td><td></td></tr></table> 1 mark for each correct row		Cause of variation			Characteristic	Genes only	Environment only	Both genes and environment	Break in the shell		✓		Length of the body			✓	Number of bands on shell	✓				1 1 1	AO2 3.6.1
	Cause of variation																							
Characteristic	Genes only	Environment only	Both genes and environment																					
Break in the shell		✓																						
Length of the body			✓																					
Number of bands on shell	✓																							

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	mutation	allow examples of agents which can cause mutation such as ionising radiation	1	AO1 3.6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	a (parent) giraffe grew a long neck the parent / giraffe passed the long neck to the offspring	allow acquired characteristic passed to offspring	1 1	AO2 3.6.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	some Abert's squirrels became isolated / separated	allow geographical isolation	1	AO2 3.6.2c
	(there was genetic) variation in the (separated) population due to (random) mutations		1	
	(and) the different environments favoured different characteristics	allow competition due to environment	1	
	caused squirrels with (different) favourable characteristics to survive and reproduce	allow natural selection	1	
	they passed on (the favourable) genes / alleles		1	
	(until) the new population of Kaibab squirrels were unable to successfully interbreed with Abert's squirrels	allow until interbreeding no longer produced fertile offspring	1	

Total		13
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	any two from: • (same) age • (same) sex • (same) room temperature • (same) incline on running machine • (same) fitness of the students	allow gender allow (same) BMI allow all non-smokers allow all non-asthmatic allow no heart / lung disease allow no caffeine ingested	2	AO4 3.2.6g 6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	zero (error)	allow systematic (error)	1	AO4 3.2.6g 6.2.4 6.1 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	any one from: • adjust the zero button • subtract 1.5 seconds to each time • count until 31.5 sec		1	AO4 3.2.6g 6.2.4 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	any one from: • use / include more students • repeat investigation (two or more times) and calculate the mean • use wider range of exercise speeds • use smaller intervals for exercise speeds • check heart rate has returned to resting heart rate after each exercise		1	AO4 3.2.6g 6.2.4 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	y-axis labelled as 'mean heart rate in beats per minute'		1	AO2 3.2.6g 6.3.9 6.3.11 6.2.4
	correct scale		1	
	all points correctly plotted	allow 1 mark for 3 correct	2	
	suitable curve of best fit	allow tolerance $\pm$ half a small square	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6	as speed (of exercise) increases so does heart rate		1	AO3 3.2.6g 6.3.12 6.2.4
	until 9.0 km/hr after which it is constant		1	
	or			
	until 147 beats per minute after which it is constant	allow descriptions of constant such as stays the same		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.7	both prevent falls	allow exercise is more effective than education talks (at preventing falls)	1	AO3 3.2.6g
	the exercise prevents 50.1% more falls	allow exercise prevents 52.8% of falls compared to 2.7% for education talks	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	(radius of diatom = 250 micrometres) = 2.5×10^{-4} (m)	allow 0.00025 (m)	1	AO2 3.1.5h 6.3.1 6.3.5 6.3.6 6.3.10 6.3.14
	(surface area) = $4 \times 3.14 \times 2.5 \times 10^{-4} \times 2.5 \times 10^{-4}$ = $7.85 \times 10^{-7} \text{ m}^2$	allow 0.000000785 m^2	1	
	(volume) = $\frac{4}{3} \times 3.14 \times 2.5 \times 10^{-4} \times 2.5 \times 10^{-4}$ = $6.5416667 \times 10^{-11} \text{ m}^3$	allow 0.000000000065416667 m^3	1	
	(SA : Vol) = $\frac{7.85 \times 10^{-7}}{6.5416667 \times 10^{-11}}$ = 1.20000×10^4	allow 1.19999×10^4 allow 1.2×10^4 allow 12000	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	the diatom has a much greater SA : Vol than the human	allow converse	1	AO3 3.1.5c, h, i, j 3.2.5c 6.3.15
	(so) gas exchange across the cell surface in diatom is (fast) enough to supply the organism		1	
	(but) in humans gases cannot diffuse (fast) enough (through the outside surface) to supply tissues with oxygen (and remove carbon dioxide)		1	
	(so) needs alveoli (in lungs) to increase the surface area.		1	

Total		10
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